

A Microparticle Swarm Optimizer for the Reconstruction of Microwave Images

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Abstract—A novel optimization technique known as the microparticle swarm optimizer (μ PSO) is proposed for high-dimensional microwave image reconstruction. With the proposed μ PSO, good optimization performance can be obtained especially for solving high-dimensional optimization problems. In addition, the proposed μ PSO requires only a small population size to outperform the standard PSO that uses a larger population size. Our simulation results on the reconstruction of the dielectric properties of normal and malignant breast tissues have shown that the μ PSO can perform quite well for this high-dimensional microwave image reconstruction problem.

Index Terms—Microparticle swarm optimizer (μ PSO), microwave imaging, noninvasive profile evaluation, optimization methods, particle swarm optimizer (PSO).

I. INTRODUCTION

RECENT advances in wireless technologies have created renewed interest on microwave imaging for performing noninvasive profile evaluation of an object under investigation (OUI), e.g., biomedical diagnosis of human physiologies, etc. [1]–[3]. The reconstruction of microwave images represents a complex inverse scattering problem that needs to be solved iteratively. That means the error between the measured data and the scattered fields computed from the trial solution is minimized at the end of each iteration, and the trial solution is then progressively adjusted towards the true profile of the OUI.

It is well-known that traditional deterministic techniques [4], [5] used for fast reconstruction of microwave images suffer from a major drawback, where the final image is highly dependent on the initial trial solution. In addition, it is often difficult to decide the adequacy of the initial trial solution for ensuring the correctness of the final solution. To overcome this obstacle, population based stochastic methods such as the genetic algorithm (GA) and particle swarm optimizer (PSO) have become attractive alternatives to reconstruct microwave images [6]–[9]. These techniques consider the imaging problem as a global optimization problem by imparting each individual within the population with its own fitness value as per the objective function defined for the problem, and reconstruct the correct image by searching

for the optimal solution through the use of either rivalry or co-operation strategies in the problem space.

In practice, to reconstruct accurate and detailed microwave images the investigation domain needs to be partitioned into many homogeneous virtual cells, which dictates the dimensionality of the problem. The reconstruction of dielectric properties within these cells would correspond to solving an optimization problem of high dimensionality. To handle such a high dimensionality, a large population size is usually required for the optimization technique. However, such a large population size would in turn increase the overall computational cost due to the increase in the number of fitness evaluations for the entire population per iteration. Hence, it would be ideal to have a technique that only requires a small population size to achieve an equal or better optimization performance as if a larger population size were used.

Currently, there is an existing optimization technique, the microgenetic algorithms (μ GA), which works based on the rivalry strategy and is used to complement the standard GA [7], [8]. For the PSO, which works on the cooperation strategy, such a microvariant is not yet available to the best of our knowledge. Thus, in this paper, we propose a novel microparticle swarm optimizer (μ PSO) that can be used to complement PSO of large population size for solving high-dimensional microwave image reconstruction problems. We will demonstrate the suitability of the proposed μ PSO for solving practical microwave image reconstruction problems by utilizing just a small population size, thus significantly reducing the number of fitness evaluations per iteration while maintaining the superior search ability and optimization performance.

The paper is organized as follows. In Section II, we describe the proposed μ PSO and include the choice of appropriate parameter values based on parametric studies. Section III compares the optimization performance of the proposed μ PSO and PSO of equal and larger population size for some well-known benchmark problems. In Section IV, the proposed μ PSO is applied to a microwave image reconstruction problem with regards to the detection of malignant tissues, and its performance is again compared with the PSO. Finally, the conclusion is presented in Section V.

II. μ PSO

The original PSO formulated based on the social interaction of biological systems was first proposed by Eberhart and Kennedy [10], [11], and later introduced to the antenna engineering community by Robinson and Rahmat-Samii [12]. However, the two main weaknesses that exist in the original PSO are:

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the problem of convergence and inability to handle high-dimensional optimization problems [13], [14]. It is well-known that the original PSO cannot guarantee its convergence and the particles can possibly converge to a location that does not even represent a local minimum [13]. Although the performance of PSO was reported to be nonsensitive to the population size [15], this, however, is not true in terms of computational cost for evaluating the fitness for the entire population of particles. Parametric studies have shown that a population size of 30 particles is suitable for solving most of the low-dimensional problems [16]. For high-dimensional problems (e.g., $N > 100$), on the other hand, the PSO performance starts to suffer due to the curse of dimensionality [17] and thus a large population size is required to explore the problem space thoroughly. Hence, for PSO in its original form, the overall computational cost could be increased drastically for solving practical high-dimensional problems due to the large population size.

To address the aforementioned issues, here we are proposing the μ PSO. Like PSO, the particles of μ PSO are multidimensional vectors that have the memory of the best solution found thus far, and the information about the global best solution is exchanged among all particles. Similar to μ GA, the μ PSO also utilizes a small population that consists of only a few particles and implements a set of restart operations after the population has converged. The convergence of population is defined as the percentage of particles that achieve a particular value of standard deviation on their locations which is less than a predefined convergence threshold.

If the population converges to a solution that is inferior or equal to the available best solution, that solution is blacklisted for future searches and all particles are prevented from converging to the same solution again. To achieve this preventative step and ensure the convergence of the PSO, two velocity update equations modified from the guaranteed convergence PSO (GCP SO) [13] are employed in μ PSO, which are given by

$$v_{j,n}(t+1) = wv_{j,n}(t) + c_1r_1[p_{j,\text{best},n} - x_{j,n}(t)] + c_2r_2[g_{\text{best},n} - x_{j,n}(t)] + \text{rep}_{j,n}(t) \quad (1)$$

$$v_{g,n}(t+1) = wv_{g,n}(t) - x_{g,n}(t) + g_{\text{best},n} + \rho(t)(1 - 2r_3) + \text{rep}_{g,n}(t) \quad (2)$$

where $v_{g,n}$ and $v_{j,n}$ refers to velocities within the n th dimension of the global best and the j th remaining particle, respectively; w is particles' inertia weight which is a constant that would either return to its initial value or increase by a small percentage β whenever the population is converged to a better or inferior solution, respectively; c_1 and c_2 are accelerating constants, and r_1, r_2 and r_3 are uniformly distributed random variables in the range of $[0,1]$; $\rho(t)$ is a scaling factor as defined in [13], and $\text{rep}_{i,n}(t)$ is the total repulsion experienced within the n th dimension of the i th particle from the blacklisted solutions.

In (1) and (2), the repulsion is used to repel particles away from blacklisted solutions. Unlike common approaches where particles are repelled immediately when they are inside a predefined space surrounding blacklisted solutions [18], we introduce a repulsion force that is computed based on the lines of well-known law of Coulomb of electrostatics. We have replaced

the electrostatic constant with the dynamic range of the problem space, D , and both the particles and blacklisted solutions are assumed to be charges of same polarity with unity magnitude. Hence, for the i th particle, the repulsion experienced from L blacklisted solutions can be described as

$$\text{rep}_i = D \sum_{l=1}^L \frac{d_{li}}{\|d_{li}\|^{m+1}} \quad (3)$$

where $d_{li} = \mathbf{x}_i - \hat{\mathbf{x}}_l$ is a vector pointing from the blacklisted solution l to the i th particle. $\mathbf{x}_i$ and $\hat{\mathbf{x}}_l$ are the position vectors of the i th particle and the blacklisted solution l , respectively. As can be seen from (3), the repulsion is inversely proportional to $\|d_{li}\|^{m+1}$, thus, the amount of repulsion experienced by the particles at a particular position can be controlled by the parameter m .

By the use of this Coulomb-like repulsive approach, two immediate advantages can be gained over other commonly used approaches [18]. The first is the removal of the overhead for calculating the suitable size of the space needed to enclose the blacklisted solutions and the amount of repulsion force needed to repel the particles, as these parameters are extremely difficult to determine for high-dimensional problems. Another advantage is the flexibility of controlling the repulsion on particles through the use of parameter m . By properly choosing the value of m , we can prevent particles from converging to the blacklisted solutions while still allowing them to explore the nearby surrounding space without experiencing too much of unnecessary repulsion. At this point it should be emphasized that although the analogy of identical charges has been used for particles and blacklisted solutions, but the repulsion should only exist between particles and blacklisted solutions and not among particles themselves.

The overall operation of the proposed μ PSO is presented in Fig. 1. It can be seen that on top of the restart operations and the computation of repulsion matrix, μ PSO has inherited some of its operations from the original PSO and GCP SO. This inheritance allows us to set μ PSO parameters such as c_1 and c_2 to values that were designed for the original PSO without worrying about their appropriateness. As for the choice of other parameters such as w, m and neighborhood topologies, the choice of these parameter values are decided based on our parametric studies, which will be discussed below.

A. Choice of Suitable Parameter w

In the original PSO, the inertia weight w has a vital role to play as to how the particles conduct their search. The magnitude of w encourages the particles to explore the space either globally or locally. To achieve a balance between global and local searches, w is usually decreased linearly from 0.9 to 0.4 throughout the whole optimization process [12], [19].

For μ PSO, however, the same approach cannot be applied due to the fact that the location and velocity matrices for all particles will be regenerated randomly after the execution of restart operations. As we cannot foresee the frequency of population convergence, and also when and where will it occur, a new approach must be adopted to set the value of w so as to achieve a balance between global and local searches.

TABLE I
LIST OF TEST FUNCTIONS

Test Functions	Expression
Rastrigin	$f_1(x) = \sum_{n=1}^N [x_n^2 - 10 \cos(2\pi x_n) + 10], -2 \leq x_n \leq 2$
Rosenbrock	$f_2(x) = \sum_{n=1}^{N-1} [100(x_{n+1} - x_n^2)^2 + (x_n - 1)^2], -2 \leq x_n \leq 2$
Griewank	$f_3(x) = 1 + \frac{1}{4000} \sum_{n=1}^N x_n^2 - \prod_{n=1}^N \cos\left(\frac{x_n}{\sqrt{n}}\right), -10 \leq x_n \leq 10$
Schwefel	$f_4(x) = 418.9829N + \sum_{n=1}^N x_n \sin\left(\sqrt{ x_n }\right), -500 \leq x_n \leq 500$
Ackley	$f_5(x) = 20 + e - 20 \exp\left(-0.2 \sqrt{\frac{1}{N} \sum_{n=1}^N x_n^2}\right) - \exp\left(\frac{1}{N} \sum_{n=1}^N \cos(2\pi x_n)\right), -2 \leq x_n \leq 2$

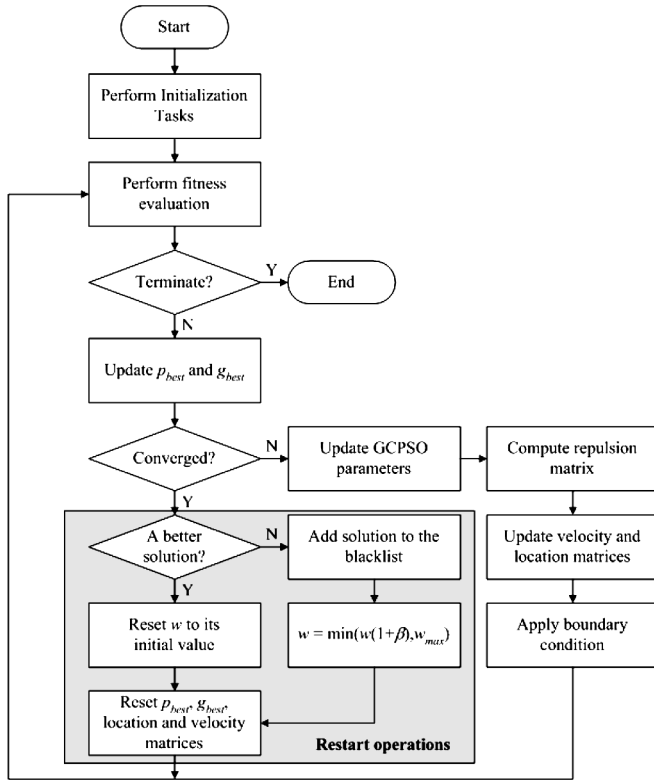


Fig. 1. Flowchart of the proposed μ PSO.

To establish a suitable strategy for setting w , we need to first consider the behavior of the particles in μ PSO. Since the parameter w controls the ability to conduct either global or local search, this information could in turn be used to control the speed of population convergence in μ PSO as there are only a small number of particles exist in the population. That is, the μ PSO population will converge faster for a small value of w and converge slowly for a large w . In addition, the inclusion of repulsion term in (1) and (2) indicates that particles have the memory about the location of blacklisted solutions. As the presence of these blacklisted solutions can only be determined and updated after the population reaches its convergence, it would be ideal for particles to start with a small value of w , w_{init} , so that undesired solutions can be discovered quickly for avoiding in later searches.

Once the choice of w is made, the next step is to determine when to increase w so as to perform a more thorough search

within the problem space. In our implementation, we have decided to increase w whenever the population is converged to an undesired solution. The amount of increase is specified by the parameter β , which is defined as a small percentage of the current value of w and a typical value for β is equal to 0.1 or 10%. If no better solution can be found in the subsequent convergence, the value of w would again increase by β until it reaches its maximum allowed value, w_{max} , which is usually set to 0.9. On the other hand, if a better solution is found in the subsequent convergence then the value of w will be reset to w_{init} .

Finally, to conclude our w -setting strategy, we need to determine a suitable value for the parameter w_{init} . To do this, we have varied the value of w from 0.2 to 0.7, in steps of 0.1. The suitability of these six alternative w_{init} values were judged based on their corresponding μ PSO performance for five well-known benchmarking test functions listed in Table I for dimensions of 100, 200, 500, and 1000.

We have repeatedly tested each case for 1000 times, and for each test case the simulation is set to terminate after the μ PSO has completed 3000 fitness evaluations. In addition, we have also considered two different population sizes of 3 and 5 particles for each test case. That is for each population size, we have a total of 20 000 test cases. Fig. 2(a) and (b) shows the histogram of percentage for each w_{init} value that achieves the best performance for the population size of 3 and 5 particles, respectively. It can be seen that in both cases, about 70% of the best performance was achieved by the choice of w_{init} values lying within the range of 0.2 to 0.4. As the population of μ PSO only consists of a few particles, based on these results, we have decided to set the value of w_{init} to 0.3.

In summary, we have adopted a dynamic approach for setting the value of w . Based on the quality of solutions found the particles can automatically adjust its search preference *viz.*, whether to continue its local searches or to gradually start to explore more globally into the problem space.

B. Choice of a Suitable Parameter m

In the proposed μ PSO, the repulsion experienced by the particles is inversely proportional to $\|d_{ii}\|^{m+1}$ and the strength of the Coulomb-like repulsion force can therefore be adjusted by the value of parameter m . In other words, for a particle at an arbitrary location, it would experience a stronger repulsion if m is a small value. Conversely, the repulsion would be weaker if a large value is chosen for m . Ideally one would like to set m to a

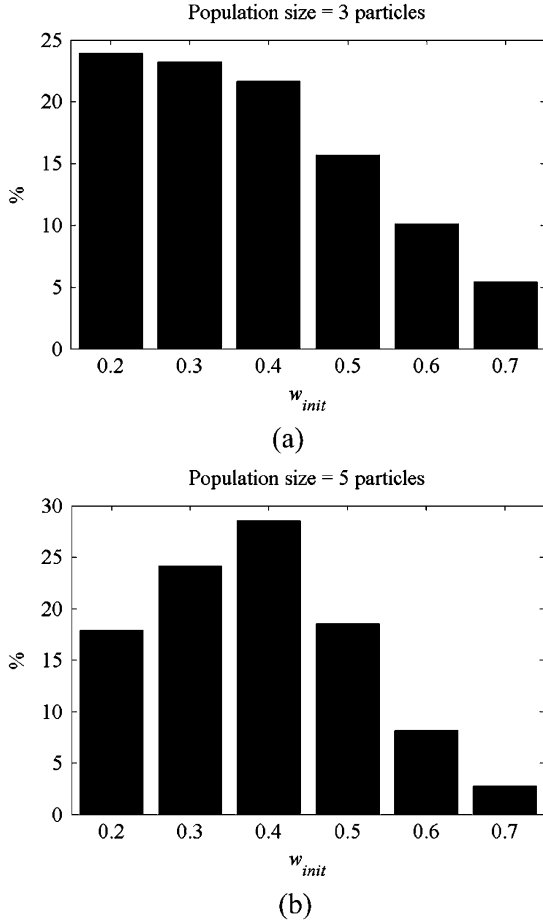


Fig. 2. Percentage of each w_{init} value achieving the best performance for the population size of: (a) 3 particles; (b) 5 particles.

value such that it can produce sufficient repulsion force to repel the particles away from blacklisted solutions while at the same time allowing them to search spaces surrounding the blacklisted solutions.

To find an appropriate value for m , we have simulated the performance for five test functions (Table I) for several values of m and compared their corresponding performances. The dimensions of these test functions have again been set to 100, 200, 500, and 1000, and each test case were repeated for 1000 times. The μ PSO is implemented with three particles as its population, and is set to terminate after the end of the 3000th fitness evaluation.

Our simulation results reveal that different values of m will not significantly vary the optimization trends for the μ PSO. But it does, however, affect the algorithm's ability of finding a solution with better fitness. For example, consider Fig. 3(a) and (b), which shows the average fitness value after 1000 simulations for the Schwefel function of dimension of 500. From Fig. 3(a), on a gross level, it can be seen that the overall optimization trend for different m values do not differ significantly. However, if we zoom in our results closely, as shown in Fig. 3(b), we can observe that as the value of m increases the fitness of the solution will also improve, note that the y axis is of the order of 10^5 . It should also be noted that although better solutions can be

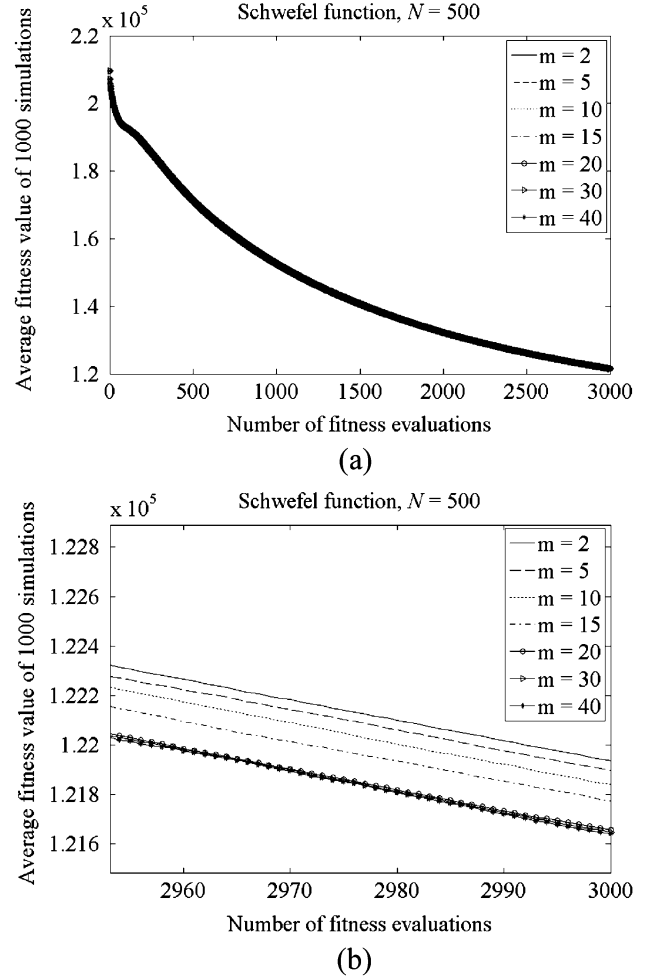


Fig. 3. Comparison of optimization performances for the Schwefel function ($N = 500$) when different m values are used. (a) Overall performance. (b) Zoomed in version.

obtained using larger m values, this improvement tends to saturate after $m = 20$. Hence, based on these observations, we have decided that $m = 20$ to be an appropriate value for the μ PSO.

C. Choice of a Suitable Neighborhood Topology

The original PSO uses a neighborhood topology known as the g_{best} topology to provide an instantaneous communication among all particles. Through this fully connected social network structure, a rapid convergence rate can be achieved. However, for this topology, it is possible that particles may converge prematurely when the position of g_{best} is not updated regularly.

To solve this problem, various kinds of neighborhood topologies have been proposed to impede the communication among particles and preserve the swarm diversity [20]. Such topologies include $lbest$ and von Neumann topologies. It has been found that with the use of these slower communication schemes, the overall accuracy of reconstructed microwave images by the PSO can be improved [21].

For the case of μ PSO, on the other hand, it is not necessary to consider neighborhood topologies other than the g_{best} topology. As the typical population size for μ PSO is only three to five

TABLE II
TYPICAL PARAMETER VALUES USED FOR IMPLEMENTING THE μ PSO

Population Size	w_{init}	w_{max}	c_1, c_2	Population Convergence	Convergence Threshold	m	β
3	0.3	0.9	1.49	80%	0.001	20	10%

TABLE III
RESULTS OBTAINED BY THE μ PSO AND PSO FOR FIVE TEST FUNCTIONS OF DIFFERENT DIMENSIONS.
THE BEST SOLUTION FOUND IN EACH TEST CASE HAS BEEN HIGHLIGHTED IN BOLD

Test functions	Dimension (N)	μ PSO 3 particles		PSO 3 particles		PSO 30 particles	
		Avg.	Std. Dev.	Avg.	Std. Dev.	Avg.	Std. Dev.
$f_1(x)$	100	150.115	19.876	605.815	48.733	476.350	86.942
	200	340.195	50.001	1,428.067	90.660	1,242.030	159.984
	500	1,598.512	237.985	3,973.413	248.750	3,714.792	386.176
	1000	5,073.618	724.380	7,924.086	533.686	7,882.382	803.430
$f_2(x)$	100	234.131	90.111	15,579.829	2,870.620	1,521.069	285.793
	200	1,202.964	394.502	36,822.684	4,374.183	7,607.021	1,031.718
	500	13,987.955	4,965.408	113,711.634	9,649.289	44,204.286	3,614.021
	1000	79,803.073	20,496.111	251,715.160	14,069.577	130,916.687	8,198.182
$f_3(x)$	100	0.057	0.034	1.379	0.044	1.034	0.040
	200	0.431	0.101	1.947	0.076	1.289	0.028
	500	1.402	0.116	3.502	0.131	2.382	0.069
	1000	3.209	0.298	6.350	0.221	4.588	0.123
$f_4(x)$	100	19,901.249	1,544.222	27,850.416	1,439.870	22,906.497	2,382.051
	200	42,258.969	1,002.630	62,472.722	1,002.142	55,303.823	1,003.942
	500	123,188.925	5,260.418	175,889.413	3,334.768	163,254.450	6,604.261
	1000	283,040.131	8,851.563	372,082.456	5,094.437	352,337.907	9,565.432
$f_5(x)$	100	3.483	0.236	4.409	0.171	2.770	0.180
	200	3.679	0.175	4.783	0.125	3.507	0.117
	500	4.137	0.138	4.954	0.083	4.170	0.075
	1000	4.430	0.204	5.062	0.065	4.499	0.057

particles, which is minutely small when compared to the standard PSO, the information about the best solution would spread throughout the whole population in a matter of just a few iterations, even when the slowest *lbest* topology is employed. In addition, with the inclusion of the repulsion term in the velocity

update equations and the execution of restart operations, the particles will be prevented from converging to the same location, thus making it capable enough to avoid premature convergence that degrades the original PSO. Thus, for the implementation of μ PSO, we only consider the *gbest* topology.

III. OPTIMIZATION PERFORMANCE BENCHMARKS

To test the effectiveness of μ PSO, test functions listed in Table I have once again been utilized to compare the optimization performances between μ PSO and the standard PSO. Since we are focusing on solving high-dimensional problems, all test functions have their dimensions set to 100, 200, 500, and 1000.

The μ PSO is implemented with a population size of just three particles and Table II lists other μ PSO parameters. The standard PSO is implemented using the description given in [16], and for the sake of comparison we have considered the PSO population sizes of 3 and 30 particles. In the implementation of μ PSO and PSO, we have used the damping boundary to enclose the problem space so that both techniques can allow a robust and consistent search for these problems of high dimensionality [22]. Finally, both μ PSO and PSO are set to terminate after 3000 fitness evaluations, and the final result is taken as the average of 1000 independent simulations.

In Table III, the final average taken over 1000 simulations and their corresponding standard deviation values are tabulated. It is clear that in most cases the μ PSO performs much better than the standard PSO of equal as well as larger population sizes. However, we must be reminded of “no free lunch theorem” which states that “for any algorithm, any elevated performance over one class of problems is exactly paid for in performance over another class” [23]. As a proof of the validity of this theorem, we can see the performance of μ PSO becomes inferior when compared to the PSO for the case of Ackley function of dimension of 100 and 200. But it is fair to say that in these two cases, the performance of the μ PSO is still better than the standard PSO of same population size. In addition, as μ PSO is designed for solving high-dimensional optimization problems, it is interesting to find that, especially for the Ackley function, even when the problem dimension increases, the performance of μ PSO will not deteriorate as much as the standard PSO and, thus, can still function as an effective optimization technique.

As a graphical comparison, Fig. 4 shows the μ PSO and PSO performances for the Griewank function for four different dimensionalities. It can be seen that the proposed μ PSO has exercised its small population of particles much more effectively than the standard PSO of same population size. The PSO with larger population size is also not able to compete with μ PSO, as it would require many more computations before it can match the solution of μ PSO. In addition, unlike the standard PSO, the search characteristics of the μ PSO can provide extra boosts for its optimization performance and this is evident in Fig. 4(a) and (b) at the fitness evaluation numbers 700 and 1260, respectively. Thus, from these results, we have demonstrated the efficiency and efficacy of μ PSO for solving high-dimensional optimization problems. Next, we will apply it to a microwave image reconstruction problem to characterize its performance for such a class of important engineering problems.

IV. APPLICATION OF μ PSO FOR THE RECONSTRUCTION OF MICROWAVE IMAGES

To investigate and compare the μ PSO performance for solving microwave image reconstruction problems, we have

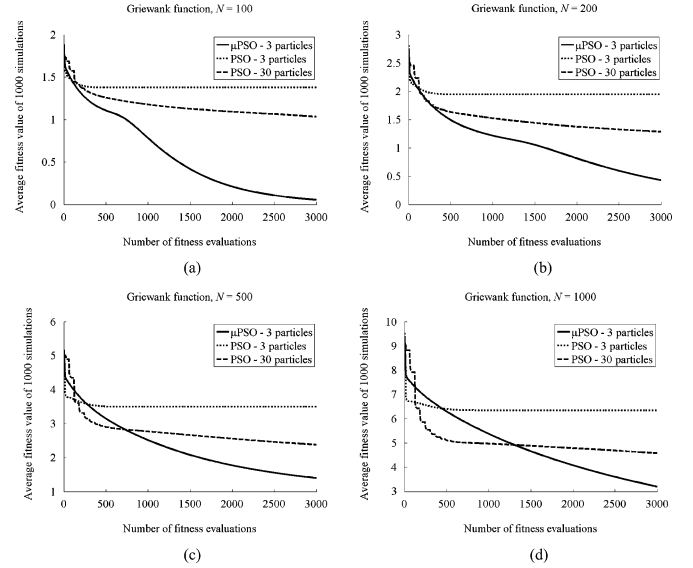


Fig. 4. The comparison of optimization performances between the μ PSO and standard PSO for the Griewank function of dimensions of: (a) 100; (b) 200; (c) 500; (d) 1000.

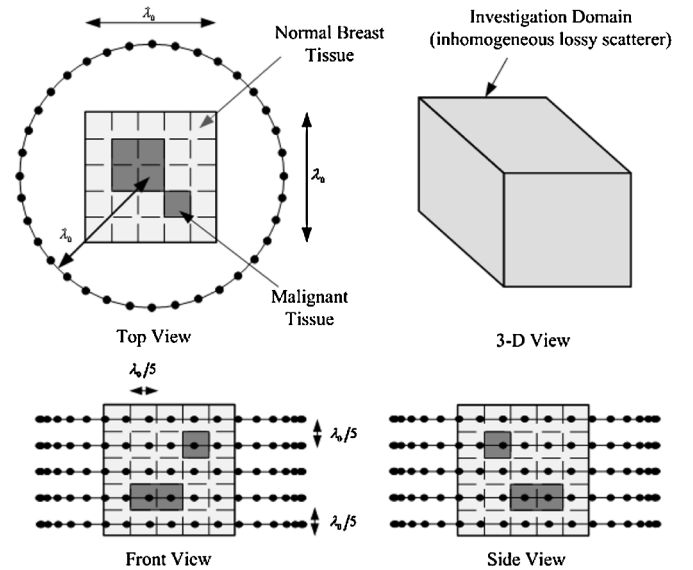


Fig. 5. The schematic of inhomogeneous lossy dielectric scatterer and the imaging system considered. The top, front, and side views are assumed to be transparent so that the location of two small malignant tissues can be revealed. The malignant tissues are represented as dark shaded cells.

adopted the single illumination, multiview approach to minimize the impact on the overall computational cost from the computation of forward solutions. Also, due to the absence of the measured data, we have synthesized our measured scattered field values using the FDTD technique and used MoM with volume integral formulation for the computation of forward solutions. By using these two different numerical techniques we have avoided committing the inverse crime [24].

For our investigation, we have considered an inhomogeneous lossy dielectric scatterer shown in Fig. 5. The scatterer is characterized by the dielectric properties of the normal breast tissue and malignant tissue at 800 MHz, where the relative permittivity

ϵ_r for the normal breast tissue is 16 and the electrical conductivity σ is 0.16 S/m. For the malignant tissue, its relative permittivity ϵ_r is 57.2 while the electrical conductivity σ is around 1.08 S/m [25], [26].

Here, we have assumed our investigation domain as the physical size of the lossy scatterer surrounded by water. The investigation domain is virtually partitioned into 125 equal sized sub-cells. Inside this lossy scatterer, we have embedded two small malignant tissues which are represented by the dark shaded cells in Fig. 5.

An imaging system that consists of five uniform circular arrays (UCAs) is used to surround the scatterer, and each UCA has 36 antenna elements that are equally spaced around the array. The radii of the UCAs are equal to λ_0 , where λ_0 is the free space wavelength of our incident plane wave of frequency $f_0 = 800$ MHz. The measurement configuration is presented in Fig. 5, where the effects of mutual coupling between the antenna elements have been neglected in our investigation. The incident plane wave is assumed to be propagating along the x axis, while the electric field vector is polarized along the z axis. The signal-to-noise ratio (SNR) in this example is assumed to be 30 dB.

In this problem, the goal is to reconstruct both ϵ_r and σ values of different tissues. Knowing that there are high-computational costs associated with the forward solutions, it will help if particles are prevented from reaching the physically invalid solutions, e.g., a tissue with ϵ_r equal to that of a normal breast tissue but σ equal to that of a malignant one, etc. To achieve this, a Gaussian distribution function is used to generate 100 samples of the normal breast tissue and 100 samples of the malignant tissue. The normal breast tissue samples are formed by combining 100 samples of ϵ_r and σ values, each with a mean of 16 and 0.16, respectively, while the standard deviation is set to 0.02. Similarly, the malignant tissue samples are formed by combining 100 samples of ϵ_r and σ , with a mean of 57.2 and 1.08, respectively, and a standard deviation of 0.005.

For the implementation, except for the population size, the μ PSO and PSO parameters are fixed based on the values given in Table II and the description in [16], respectively. The μ PSO is populated using five particles while the PSO is populated using 25 particles. Each particle is a vector of dimension of 125. For each dimension, the particles search for the appropriate tissue index, which corresponds to a set of valid ϵ_r and σ values. Hence, the search range of each dimension can be defined as $x_{i,n} \in [1, 200]$, where $1 \leq i \leq 5$ for the μ PSO, $1 \leq i \leq 25$ for the PSO, and $1 \leq n \leq 125$ for both techniques. Both μ PSO and PSO are linked to the problem via an objective function defined as the normalized root mean square error (RMSE) between the measured and computed values of the scattered field $\vec{E}^s$, i.e.

$$OF = \sqrt{\frac{1}{V} \sum_{v=1}^V \frac{|\vec{E}_{\text{meas}}^s(\rho_v, \phi_v, z_v) - \vec{E}_{\text{comp}}^s(\rho_v, \phi_v, z_v)|^2}{|\vec{E}_{\text{meas}}^s(\rho_v, \phi_v, z_v)|^2}} \quad (4)$$

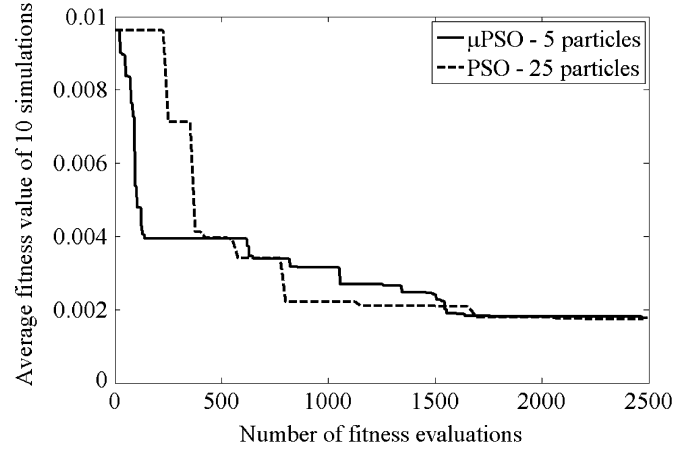


Fig. 6. Comparison of optimization performances for the microwave imaging example.

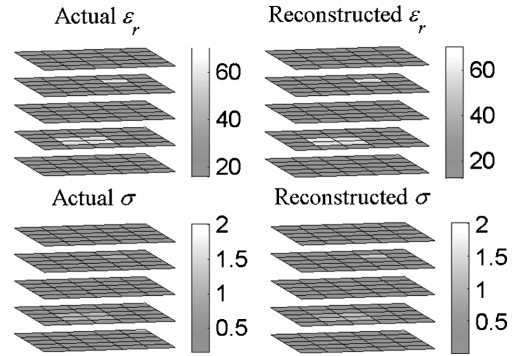


Fig. 7. Comparison of the actual distributions of ϵ_r and σ values inside the lossy dielectric scatterer and the final image reconstructed by the μ PSO.

where V is the total number of receivers used in the imaging system and $\vec{E}_{\text{meas}}^s(\rho_v, \phi_v, z_v)$, $\vec{E}_{\text{comp}}^s(\rho_v, \phi_v, z_v)$ are the measured and computed values of $\vec{E}^s$ at location (ρ_v, ϕ_v, z_v) , respectively.

Our final result is arrived after performing 10 independent simulation runs, each with 500 iterations for the μ PSO and 100 iterations for the PSO. This termination criterion is set for the purpose of having a fair comparison between the two techniques, as their performances will be compared in terms of number of fitness evaluations. Fig. 6 shows the comparison of optimization performances between the μ PSO and PSO. As expected, it can be seen that the μ PSO has started with a much rapid convergence rate than the PSO. Also seen in Fig. 6, the similarities in the fitness of the final solutions need to be interpreted carefully to avoid any wrong impression about the suitability of applying μ PSO for microwave image reconstruction problems. Since each imaging problem space is unique and unknown, the μ PSO may not outperform the PSO in all the cases [23]. Thus, the similarity in the results shown in Fig. 6 can only reflect one such isolated case, i.e., the problem scenario shown in Fig. 5 where the problem dimension is equal to 125 due to the computational constraints. However, with the

rapid initial convergence rate and other characteristics such as the restart scheme built into the algorithm, it would still be more suitable to use μ PSO for solving microwave image reconstruction problems, as it is more likely to find a better solution than the PSO for problems of higher dimensions. Fig. 7 shows the comparison between the actual distributions of ε_r and σ inside the investigation domain and the final reconstructed image by the μ PSO. As seen in Fig. 7, although the contrast in σ values is not as large as ε_r values, the location of the two small malignant tissues can still be easily identified. Thus, these images have shown that the μ PSO can perform well for microwave image reconstruction problems.

V. CONCLUSION

In this paper, we have proposed a novel optimization technique for solving high-dimensional optimization problems. The performance of the proposed μ PSO has been conformed against various well-known test functions. The advantage of the proposed μ PSO arises from having a small population size, which reduces drastically the computational cost associated with fitness evaluation as against the standard PSO of large population size. In addition, even with its small population size, our simulation results have shown that the proposed μ PSO is more suitable than the PSO of a larger population size for solving high-dimensional optimization problems.

We have also applied the proposed μ PSO to a microwave image reconstruction problem, and reconstructed a satisfactory image for the OUI. Although we have only considered an image reconstruction problem of dimension of 125 due to the limitation in the available computational resources for computing forward solutions, it is expected that μ PSO would continue to offer outstanding performance for problems of higher dimensions.

Finally, there is scope for improving the proposed μ PSO by including novel aspects such as the ensemble method [27] and other hybrid techniques, etc. It is our hope that the proposed μ PSO will help to lead to novel approaches for solving complex EM optimization problems.

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